

DSA0602 – Data Handling and Visualization

Homework for Unit II – BrightPath Quiz Score Analysis

Dataset: brightpath_quiz_scores.csv (160 rows, 5 columns)

Date: 29/07/2026

Scenario 2

Python Basics: Distribution Shape

PYTHON

Code & Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

df = pd.read_csv("brihtpath_quiz_scores.csv")
pb = df[df['course'] == 'Python Basics']['quiz_score']

# 1. Histogram - two different bin widths side by side
fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
axes[0].hist(pb, bins=6, color='steelblue', edgecolor='black')
axes[0].set_title("Python Basics Quiz Scores (6 bins)")
axes[0].set_xlabel("Quiz Score"); axes[0].set_ylabel("Frequency")

axes[1].hist(pb, bins=16, color='darkorange', edgecolor='black')
axes[1].set_title("Python Basics Quiz Scores (16 bins)")
axes[1].set_xlabel("Quiz Score"); axes[1].set_ylabel("Frequency")
plt.tight_layout()
plt.show()
# With only 6 bins the shape looks like a fairly smooth single hump. At 16 bins,
# small gaps and bumps appear (since each bin now holds very few of the 40
# students), which can look like extra "peaks" that are really just sampling
# noise rather than genuine structure in the data.

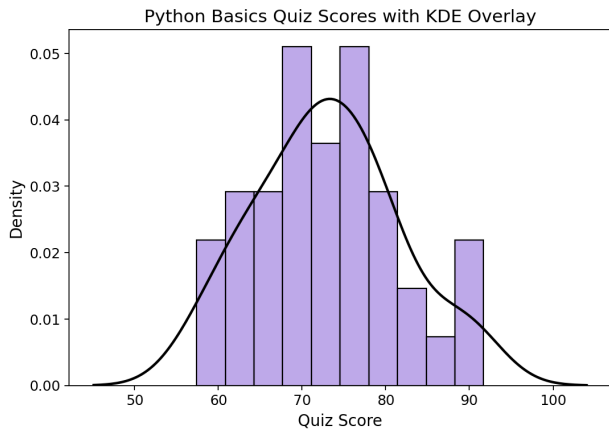
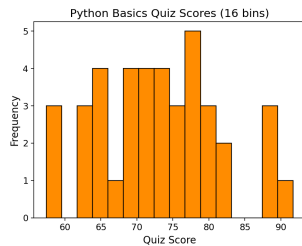
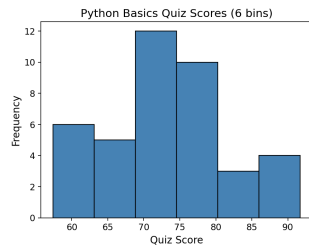
# 2. Histogram with a smoothed density (KDE) curve overlaid
plt.figure(figsize=(7, 5))
sns.histplot(pb, bins=10, stat='density', color='mediumpurple', edgecolor='black', alpha=0.6)
sns.kdeplot(pb, color='black', linewidth=2)
plt.title("Python Basics Quiz Scores with KDE Overlay")
plt.xlabel("Quiz Score"); plt.ylabel("Density")
plt.tight_layout()
plt.show()

# 3. Mean, median, standard deviation
mean_val = pb.mean()
median_val = pb.median()
std_val = pb.std()
print(f"Mean: {mean_val:.2f}")
print(f"Median: {median_val:.2f}")
print(f"Std: {std_val:.2f}")
# Mean = 73.04, Median = 72.90, Std = 8.58.
# Mean and median are almost identical here, which is itself evidence the
# distribution is close to symmetric. Since there are no extreme outliers
# pulling the mean away from the bulk of the data (see point 5 below), the
# MEAN is a fair and reasonably interpretable measure of center for this
# course - it isn't being distorted by a handful of extreme scores the way
# it would be in a skewed distribution.

"""
4. Shape of the distribution:
The histogram is roughly unimodal (one central hump) and close to
symmetric - scores cluster in the 65-80 range with roughly similar tails
falling off on either side toward the 55-60 low end and the 85-90 high
end. It doesn't show the sharp one-sided tail of a skewed distribution,
nor does it show two separated humps (bimodal) - it looks like a single,
moderately-wide "typical" spread of exam performance.

5. Outliers:
Using the standard 1.5xIQR rule, there are NO flagged outliers for Python
Basics - the lowest score (57.4) and highest score (91.7) both fall
comfortably within the whisker range. Visually, the histogram tails taper
off gradually rather than showing an isolated bar sitting apart from the
rest of the distribution, which agrees with that conclusion.
"""
```

Output



R

Code & Analysis

```
library(ggplot2)
library(gridExtra)
library(dplyr)

df <- read.csv("brightpath_quiz_scores.csv")
pb <- df %>% filter(course == "Python Basics")

# 1. Histogram - two different bin widths side by side
p1 <- ggplot(pb, aes(x=quiz_score)) +
  geom_histogram(bins=6, fill="steelblue", color="black") +
  labs(title="Python Basics Quiz Scores (6 bins)", x="Quiz Score", y="Frequency") +
  theme_minimal()

p2 <- ggplot(pb, aes(x=quiz_score)) +
  geom_histogram(bins=16, fill="darkorange", color="black") +
  labs(title="Python Basics Quiz Scores (16 bins)", x="Quiz Score", y="Frequency") +
  theme_minimal()

grid.arrange(p1, p2, ncol=2)
# With only 6 bins the shape looks like a fairly smooth single hump. At 16 bins,
# small gaps and bumps appear (since each bin now holds very few of the 40
# students), which can look like extra "peaks" that are really just sampling
# noise rather than genuine structure in the data.

# 2. Histogram with a smoothed density (KDE) curve overlaid
p3 <- ggplot(pb, aes(x=quiz_score)) +
  geom_histogram(aes(y=after_stat(density)), bins=10, fill="mediumpurple",
    color="black", alpha=0.6) +
  geom_density(color="black", linewidth=1) +
  labs(title="Python Basics Quiz Scores with KDE Overlay", x="Quiz Score", y="Density") +
  theme_minimal()
print(p3)

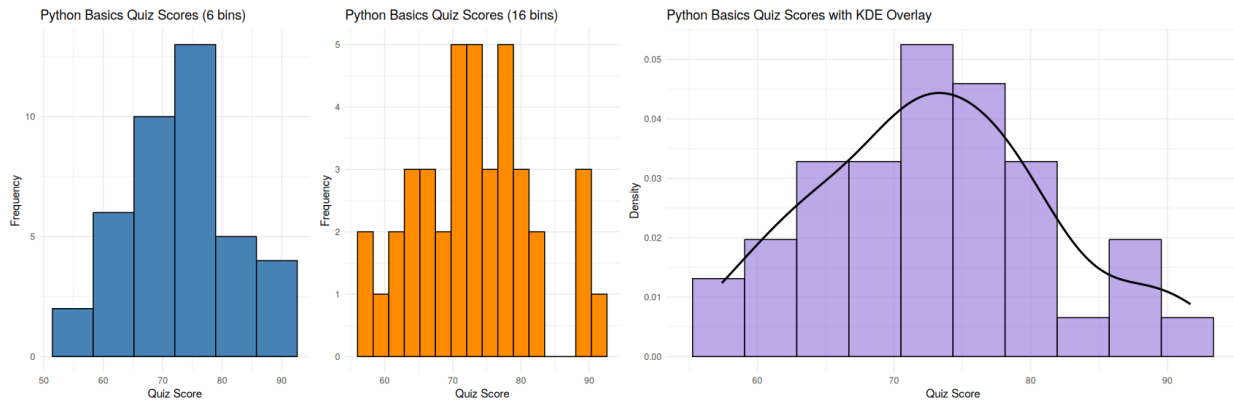
# 3. Mean, median, standard deviation
mean_val <- mean(pb$quiz_score)
median_val <- median(pb$quiz_score)
std_val <- sd(pb$quiz_score)
cat(sprintf("Mean: %.2f\n", mean_val))
cat(sprintf("Median: %.2f\n", median_val))
cat(sprintf("Std: %.2f\n", std_val))
# Mean = 73.04, Median = 72.90, Std = 8.58.
# Mean and median are almost identical here, which is itself evidence the
# distribution is close to symmetric. Since there are no extreme outliers
# pulling the mean away from the bulk of the data (see point 5 below), the
# MEAN is a fair and reasonably interpretable measure of center for this
# course - it isn't being distorted by a handful of extreme scores the way
# it would be in a skewed distribution.

# 4. Shape of the distribution:
# The histogram is roughly unimodal (one central hump) and close to
# symmetric - scores cluster in the 65-80 range with roughly similar tails
# falling off on either side toward the 55-60 low end and the 85-90 high
# end. It doesn't show the sharp one-sided tail of a skewed distribution,
# nor does it show two separated humps (bimodal) - it looks like a single,
# moderately-wide "typical" spread of exam performance.

# 5. Outliers:
# Using the standard 1.5xIQR rule, there are NO flagged outliers for Python
```

```
# Basics - the lowest score (57.4) and highest score (91.7) both fall
# comfortably within the whisker range. Visually, the histogram tails taper
# off gradually rather than showing an isolated bar sitting apart from the
# rest of the distribution, which agrees with that conclusion.
```

Output



Scenario 3

Comparing All Four Courses

PYTHON

Code & Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

df = pd.read_csv("brihtpath_quiz_scores.csv")
course_order = ['Python Basics', 'Data Science', 'Web Development', 'Digital Marketing']

# 1. Overlaid density plot - all four courses on the same axes
plt.figure(figsize=(8, 5.5))
for course in course_order:
    sns.kdeplot(df[df['course'] == course]['quiz_score'], label=course, fill=True, alpha=0.25)
plt.title("Quiz Score Density by Course (Overlaid)")
plt.xlabel("Quiz Score"); plt.ylabel("Density")
plt.legend(title="Course")
plt.tight_layout()
plt.show()

# 2. Faceted small-multiples - one histogram per course, shared x-axis
g = sns.FacetGrid(df, col='course', col_order=course_order, col_wrap=2,
                  sharex=True, sharey=True, height=3.2, aspect=1.3)
g.map(plt.hist, 'quiz_score', bins=10, color='steelblue', edgecolor='black')
g.set_axis_labels("Quiz Score", "Frequency")
g.set_titles("{col_name}")
plt.tight_layout()
plt.show()
```

"""

3. Comparing the four distributions (using actual computed statistics):

Course	Mean	Std	IQR
Python Basics	73.04	8.58	11.38
Data Science	70.44	10.07	12.88
Web Development	71.96	15.41	26.98
Digital Marketing	78.72	10.08	6.02

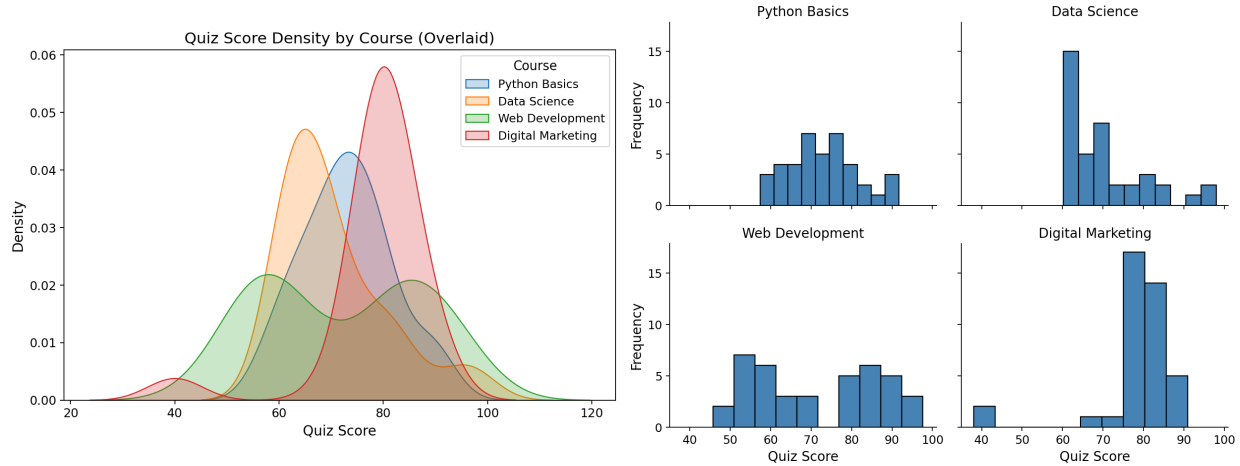
- Most spread out: Web Development, by a wide margin (std ~15.4, IQR ~27 - more than double every other course's IQR).
- Most tightly clustered: Digital Marketing (IQR only ~6.0, the tightest of all four), even though its standard deviation (~10.08) looks similar to Data Science's - the IQR tells a different, more central story because Digital Marketing's spread comes from two extreme low scorers, not from the bulk of the class being spread out.
- Bimodal / multimodal: Web Development. Its histogram shows two visible clusters - roughly 45-65 and again 72-97 - with a noticeably thin gap around 65-72. In a real self-paced platform, this pattern often shows up when a course has a genuine split in student readiness: students who already had some programming/tooling background move through the material quickly and score well, while students starting from scratch struggle with the same self-paced material and fall into a lower cluster - i.e. two different populations rather than one continuum.
- Skewed courses: Data Science is right-skewed (skew ~+1.32) - most students cluster in the 60-70 range with a smaller number of high scorers (up to 98) stretching the upper tail. Digital Marketing is left-skewed (skew ~-2.92) - most students score strongly (75-90) with a couple of very low scores (38, 42) dragging a long lower tail.

4. Overlay vs facet - what each shows more easily:

The OVERLAID plot makes it easy to directly compare where each course's peak sits relative to the others on one shared scale - e.g. it's immediately visible that Digital Marketing's peak sits furthest to the right (highest typical score) while Data Science's sits furthest left. The FACETED plot makes it much easier to see each course's individual shape without visual interference from the other three overlapping colors - the Web Development panel's two-cluster (bimodal) structure is far clearer alone than when it's layered underneath three other semi-transparent curves in the overlay.

"""

Output



R

Code & Analysis

```
library(ggplot2)
library(dplyr)

df <- read.csv("brightpath_quiz_scores.csv")
course_order <- c('Python Basics', 'Data Science', 'Web Development', 'Digital Marketing')
df$course <- factor(df$course, levels=course_order)

# 1. Overlaid density plot - all four courses on the same axes
p1 <- ggplot(df, aes(x=quiz_score, fill=course)) +
  geom_density(alpha=0.3) +
  labs(title="Quiz Score Density by Course (Overlaid)", x="Quiz Score", y="Density", fill="Course") +
  theme_minimal()
print(p1)

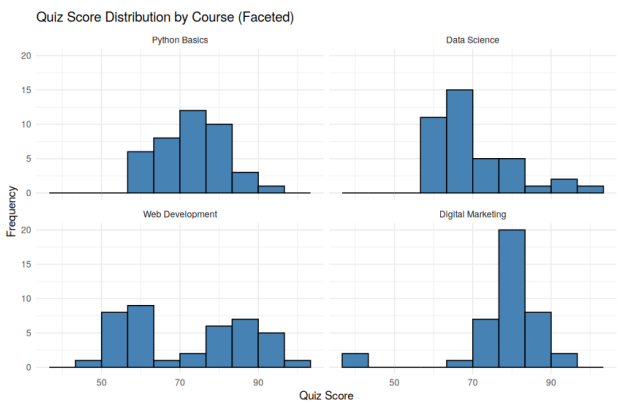
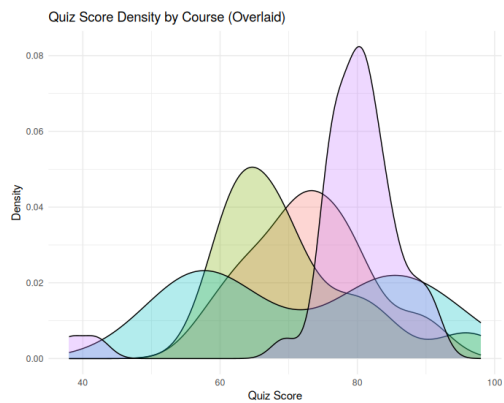
# 2. Faceted small-multiples - one histogram per course, shared x-axis
p2 <- ggplot(df, aes(x=quiz_score)) +
  geom_histogram(bins=10, fill="steelblue", color="black") +
  facet_wrap(~course, ncol=2, scales="fixed") +
  labs(title="Quiz Score Distribution by Course (Faceted)", x="Quiz Score", y="Frequency") +
  theme_minimal()
print(p2)

# 3. Comparing the four distributions (using actual computed statistics):
#
#   Course      Mean   Std   IQR
#   Python Basics  73.04  8.58  11.38
#   Data Science   70.44 10.07  12.88
#   Web Development 71.96 15.41  26.98
#   Digital Marketing 78.72 10.08   6.02
#
# - Most spread out: Web Development, by a wide margin (std ~15.4, IQR ~27 -
#   more than double every other course's IQR).
# - Most tightly clustered: Digital Marketing (IQR only ~6.0, the tightest
#   of all four), even though its standard deviation (~10.08) looks similar
#   to Data Science's - the IQR tells a different, more central story
#   because Digital Marketing's spread comes from two extreme low scorers,
#   not from the bulk of the class being spread out.
# - Bimodal / multimodal: Web Development. Its histogram shows two visible
#   clusters - roughly 45-65 and again 72-97 - with a noticeably thin gap
#   around 65-72. In a real self-paced platform, this pattern often shows
#   up when a course has a genuine split in student readiness: students
#   who already had some programming/tooling background move through the
#   material quickly and score well, while students starting from scratch
#   struggle with the same self-paced material and fall into a lower
#   cluster - i.e. two different populations rather than one continuum.
# - Skewed courses: Data Science is right-skewed (skew ~+1.32) - most
#   students cluster in the 60-70 range with a smaller number of high
#   scorers (up to 98) stretching the upper tail. Digital Marketing is
#   left-skewed (skew ~-2.92) - most students score strongly (75-90) with
#   a couple of very low scores (38, 42) dragging a long lower tail.

# 4. Overlay vs facet - what each shows more easily:
# The OVERLAID plot makes it easy to directly compare where each course's
# peak sits relative to the others on one shared scale - e.g. it's
# immediately visible that Digital Marketing's peak sits furthest to the
# right (highest typical score) while Data Science's sits furthest left.
# The FACETED plot makes it much easier to see each course's individual
```

```
# shape without visual interference from the other three overlapping
# colors - the Web Development panel's two-cluster (bimodal) structure is
# far clearer alone than when it's layered underneath three other
# semi-transparent curves in the overlay.
```

Output



Scenario 5

Boxplots and Redesign Recommendation

PYTHON

Code & Analysis

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

df = pd.read_csv("brightpath_quiz_scores.csv")
course_order = ['Python Basics', 'Data Science', 'Web Development', 'Digital Marketing']

# 1. Side-by-side boxplots - one box per course
plt.figure(figsize=(8, 5.5))
sns.boxplot(data=df, x='course', y='quiz_score', order=course_order,
            hue='course', palette='Set2', legend=False)
plt.title("Quiz Score by Course (Boxplot)")
plt.xlabel("Course"); plt.ylabel("Quiz Score")
plt.tight_layout()
plt.show()

# 2. Median, IQR, and outliers per course (computed directly from the data)
print(f"{'Course':&#x20;}{ 'Median':&#x20;}{ 'Q1':&#x20;}{ 'Q3':&#x20;}{ 'IQR':&#x20;}")
summary = {}
for course in course_order:
    sub = df[df['course'] == course]['quiz_score']
    q1, median, q3 = sub.quantile([0.25, 0.5, 0.75])
    iqr = q3 - q1
    summary[course] = iqr
    print(f"{'course':&#x20;}{ 'median':&#x20;}{ 'q1':&#x20;}{ 'q3':&#x20;}{ 'iqr':&#x20;}")

    lower, upper = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    outliers = df[(df['course'] == course) & ((df['quiz_score'] &#x26;#x20; lower) | (df['quiz_score'] &#x26;#x20; upper))]
    if len(outliers):
        print(f"    outliers -&#x26;#x20; {outliers[['student_id', 'quiz_score']].to_dict('records')}")
    else:
        print("    outliers -&#x26;#x20; none")

"""
Median / IQR / outliers per course (from the printed output above):
- Python Basics:    median 72.90, IQR 11.38, no flagged outliers
- Data Science:     median 67.65, IQR 12.88, 2 HIGH outliers
                    (BP2000: 96.0, BP2001: 98.0)
- Web Development:  median 70.85, IQR 26.98, no flagged outliers
                    (spread is continuous, not driven by isolated points)
- Digital Marketing: median 80.10, IQR 6.02, 2 LOW outliers
                    (BP4000: 38.0, BP4001: 42.0)

3. Ranking most consistent -&#x26;#x20; least consistent, by IQR/whisker length
   (NOT by mean):
   1. Digital Marketing (IQR 6.02)    - most consistent
   2. Python Basics    (IQR 11.38)
   3. Data Science      (IQR 12.88)
   4. Web Development   (IQR 26.98) - least consistent, by a large margin
"""

# 4. Investigate a flagged outlier's row (Digital Marketing's low scorers)
outlier_rows = df[df['student_id'].isin(['BP4000', 'BP4001'])]
print(outlier_rows[['student_id', 'course', 'quiz_score', 'hours_studied', 'attendance_pct']])
"""
BP4000 (score 38.0): hours_studied = 6.8, attendance_pct = 84.1
BP4001 (score 42.0): hours_studied = 10.2, attendance_pct = 79.8

Does the extra info help explain the outlier?
Partially. Both students had reasonably strong attendance (~80-84%), so
simply "not showing up" doesn't explain the very low scores. Their
hours_studied were on the lower side compared to the Digital Marketing
course's other students, but not dramatically so - other students in the
40-90% attendance range studied comparable hours and still scored 75+.
This suggests attendance and hours studied alone don't fully explain these
two outliers; something else (e.g. the quiz not matching what was covered
live, personal circumstances, or genuine content-engagement gaps not
captured by attendance/hours) is more likely at play, and is worth a direct
follow-up with these two students rather than assuming disengagement.
"""

# 5. Optional extension: boxplots split by attendance (above/below 80%)
df['AttendanceGroup'] = df['attendance_pct'].apply(lambda x: '&#x26;#x20;=80%' if x &#x26;#x20;= 80 else '&#x26;#x20;&#x26;#x20;80%')
plt.figure(figsize=(9, 5.5))
sns.boxplot(data=df, x='course', y='quiz_score', hue='AttendanceGroup',
```



```

        order=course_order, palette='Set3')
plt.title("Quiz Score by Course, Split by Attendance (Above/Below 80%)")
plt.xlabel("Course"); plt.ylabel("Quiz Score")
plt.legend(title="Attendance")
plt.tight_layout()
plt.show()
"""

Does attendance appear related to score consistency?
Looking at the split boxplots, the >=80% attendance group is generally
tighter and centered somewhat higher across most courses, while the <80%
group tends to show a wider spread and a lower median in courses like Web
Development. This suggests attendance is associated with more consistent
(and somewhat higher) outcomes, though it does not fully explain Web
Development's overall wide spread - even within its own >=80% attendance
group, scores remain fairly spread out, reinforcing that Web Development's
inconsistency likely reflects a genuine mixed-skill-level student
population rather than attendance alone.
"""

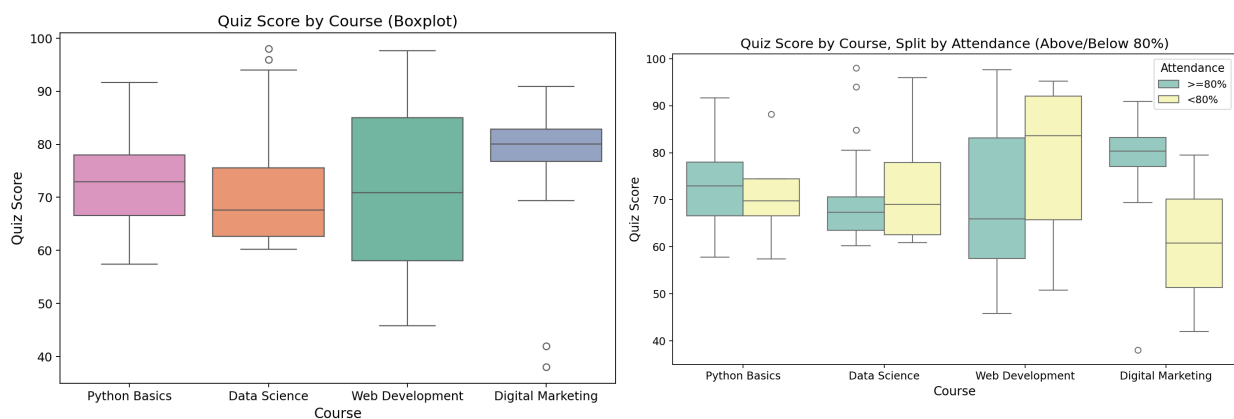
```

```

"""
FINAL RECOMMENDATION (one paragraph):
Web Development is the course most in need of redesign. Its boxplot shows
the widest interquartile range of any course (IQR = 26.98, more than double
every other course) and a median (70.85) that sits in the middle of an
unusually long box and whisker span rather than a tight cluster - meaning
students are not converging on a shared level of mastery the way they do in
Digital Marketing (IQR 6.02) or Python Basics (IQR 11.38). This isn't driven
by a couple of isolated outlier students (no points are individually flagged
by the 1.5xIQR rule); rather the spread is continuous and, combined with the
clearly bimodal histogram shape seen in Scenario 3, points to two distinct
sub-populations of students moving through the same self-paced material at
very different levels of readiness. A worthwhile next step is to investigate
whether Web Development's self-paced structure assumes prior tooling/setup
knowledge (e.g. command line, Git, local dev environment) that only some
students already have - mixed prior skill level, rather than course
difficulty alone, is the most likely explanation worth testing before any
content rewrite.
"""

```

Output



R

Code & Analysis

```

library(ggplot2)
library(dplyr)

df <- read.csv("brightpath_quiz_scores.csv")
course_order <- c('Python Basics', 'Data Science', 'Web Development', 'Digital Marketing')
df$course <- factor(df$course, levels=course_order)

# 1. Side-by-side boxplots - one box per course
p1 <- ggplot(df, aes(x=course, y=quiz_score, fill=course)) +
  geom_boxplot(show.legend=FALSE) +
  labs(title="Quiz Score by Course (Boxplot)", x="Course", y="Quiz Score") +
  theme_minimal()
print(p1)

# 2. Median, IQR, and outliers per course
for (course in course_order) {
  sub <- df %>% filter(course == !!course)

```

```

q <- quantile(sub$quiz_score, c(0.25, 0.5, 0.75))
iqr <- q[3] - q[1]
cat(sprintf("%-20s median=%.2f Q1=%.2f Q3=%.2f IQR=%.2f\n",
           course, q[2], q[1], q[3], iqr))
lower <- q[1] - 1.5 * iqr
upper <- q[3] + 1.5 * iqr
outliers <- sub %>% filter(quiz_score < lower | quiz_score > upper)
if (nrow(outliers) > 0) {
  print(outliers[, c('student_id', 'quiz_score')])
} else {
  cat("    outliers -> none\n")
}
}

# Median / IQR / outliers per course (from the printed output above):
# - Python Basics:    median 72.90, IQR 11.38, no flagged outliers
# - Data Science:     median 67.65, IQR 12.88, 2 HIGH outliers
#                     (BP2000: 96.0, BP2001: 98.0)
# - Web Development:  median 70.85, IQR 26.98, no flagged outliers
#                     (spread is continuous, not driven by isolated points)
# - Digital Marketing: median 80.10, IQR 6.02, 2 LOW outliers
#                     (BP4000: 38.0, BP4001: 42.0)

# 3. Ranking most consistent -> least consistent, by IQR/whisker length
# (NOT by mean):
# 1. Digital Marketing (IQR 6.02) - most consistent
# 2. Python Basics (IQR 11.38)
# 3. Data Science (IQR 12.88)
# 4. Web Development (IQR 26.98) - least consistent, by a large margin

# 4. Investigate a flagged outlier's row (Digital Marketing's low scorers)
outlier_rows <- df %>% filter(student_id %in% c('BP4000', 'BP4001'))
print(outlier_rows[, c('student_id', 'course', 'quiz_score', 'hours_studied', 'attendance_pct')])

# BP4000 (score 38.0): hours_studied = 6.8, attendance_pct = 84.1
# BP4001 (score 42.0): hours_studied = 10.2, attendance_pct = 79.8
#
# Does the extra info help explain the outlier?
# Partially. Both students had reasonably strong attendance (~80-84%), so
# simply "not showing up" doesn't explain the very low scores. Their
# hours_studied were on the lower side compared to the Digital Marketing
# course's other students, but not dramatically so - other students in the
# 40-90% attendance range studied comparable hours and still scored 75+.
# This suggests attendance and hours studied alone don't fully explain these
# two outliers; something else (e.g. the quiz not matching what was covered
# live, personal circumstances, or genuine content-engagement gaps not
# captured by attendance/hours) is more likely at play, and is worth a direct
# follow-up with these two students rather than assuming disengagement.

# 5. Optional extension: boxplots split by attendance (above/below 80%)
df$AttendanceGroup <- ifelse(df$attendance_pct >= 80, ">=80%", "<80%")
p2 <- ggplot(df, aes(x=course, y=quiz_score, fill=AttendanceGroup)) +
  geom_boxplot() +
  labs(title="Quiz Score by Course, Split by Attendance (Above/Below 80%)",
       x="Course", y="Quiz Score", fill="Attendance") +
  theme_minimal()
print(p2)

# Does attendance appear related to score consistency?
# Looking at the split boxplots, the >=80% attendance group is generally
# tighter and centered somewhat higher across most courses, while the <80%
# group tends to show a wider spread and a lower median in courses like Web
# Development. This suggests attendance is associated with more consistent
# (and somewhat higher) outcomes, though it does not fully explain Web
# Development's overall wide spread - even within its own >=80% attendance
# group, scores remain fairly spread out, reinforcing that Web Development's
# inconsistency likely reflects a genuine mixed-skill-level student
# population rather than attendance alone.

# FINAL RECOMMENDATION (one paragraph):
# Web Development is the course most in need of redesign. Its boxplot shows
# the widest interquartile range of any course (IQR = 26.98, more than double
# every other course) and a median (70.85) that sits in the middle of an
# unusually long box and whisker span rather than a tight cluster - meaning
# students are not converging on a shared level of mastery the way they do in
# Digital Marketing (IQR 6.02) or Python Basics (IQR 11.38). This isn't driven
# by a couple of isolated outlier students (no points are individually flagged
# by the 1.5xIQR rule); rather the spread is continuous and, combined with the
# clearly bimodal histogram shape seen in Scenario 3, points to two distinct
# sub-populations of students moving through the same self-paced material at
# very different levels of readiness. A worthwhile next step is to investigate
# whether Web Development's self-paced structure assumes prior tooling/setup
# knowledge (e.g. command line, Git, local dev environment) that only some
# students already have - mixed prior skill level, rather than course
# difficulty alone, is the most likely explanation worth testing before any
# content rewrite.

```

Output

